

M04 PROTECTED REVIEW / REPAIR

Attempt-gated targeted repair | 16 matching sections

PROTECTED: open only the matching section after a genuine independent attempt is saved. Never browse future sections as answer material. Repair is not mastery; a changed retest with this PDF closed is required when assigned.

R01 | DE04-01 | Python setup and first program

Strong evidence shows

Expected output appears; the error is preserved in your notes; the repaired file runs after closing and reopening the interpreter/editor.

Targeted repair direction

Read the final error line; inspect the smallest syntax region; rerun from a clean start.

Do not do

Do not copy a finished solution over the failed attempt. Do not erase the failed attempt. Do not mark mastery while this Review is open.

Assigned changed retry

Create a fresh file that prints a different label and calculates $9 * 6$. Introduce a different syntax error from your first attempt, explain it, then repair it without opening Review R01.

Close this Review now. Complete the changed retry independently and save both attempt histories.

R02 | DE04-02 | Variables, data types and operators

Strong evidence shows

Types match business meaning; arithmetic is applied only to measures; comparison output is boolean; you can explain why an identifier remains text.

Targeted repair direction

Print value + type. Separate identifiers from measures. Rebuild one operator/comparison at a time.

Do not do

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Assigned changed retry

Use shipment_id="S009", cartons=7, unit_cost=425.5, delivered=False. Calculate cost and `needs\_review = (cost > 2500 and not delivered)`.

Close this Review now. Complete the changed retry independently and save both attempt histories.

R03 | DE04-03 | Strings and formatting

Strong evidence shows

Whitespace changes are visible; casing is intentional; original can still be compared; f-string output matches the required label exactly.

Targeted repair direction

Use repr(raw) and repr(clean). Preserve raw text and question automatic casing when business meaning matters.

Do not do

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Assigned changed retry

Normalize " west DEPOT " to an agreed label and create a separate message containing the original and normalized values.

Close this Review now. Complete the changed retry independently and save both attempt histories.

R04 | DE04-04 | Conditions and boolean logic

Strong evidence shows

Boundary cases behave exactly as the written rule; a high Pending amount does not become priority.

Targeted repair direction

Write the English rule first. Split each boolean clause. Test just-below/exact/above boundaries.

Do not do

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Assigned changed retry

Classify a shipment as `escalate` only if `delayed_days >= 2` AND `priority_customer` is True. Test 1/2/3 days and both customer flags.

Close this Review now. Complete the changed retry independently and save both attempt histories.

R05 | DE04-05 | Lists, tuples, sets and dictionaries

Strong evidence shows

Choices match the requirement; duplicates in the set collapse intentionally; dictionary keys are meaningful/unique.

Targeted repair direction

Choose by requirement: ordered mutable sequence=list; fixed record=tuple; unique membership=set; key-value=dict.

Do not do

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Assigned changed retry

Represent warehouse zones and supervisors, a unique set of failed batch IDs, a processing sequence, and a fixed `(year, month)` period.

Close this Review now. Complete the changed retry independently and save both attempt histories.

R06 | DE04-06 | Loops and iteration

Strong evidence shows

Each input is processed once; accumulator starts at the correct value; control total matches; output list contains only eligible values.

Targeted repair direction

Initialize state before the loop, update once per eligible item, then reconcile to an independent control.

Do not do

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Assigned changed retry

From [925, 2100, 1750, 600, 2400, 1100], calculate total, count ≥ 1800 and collect values under 1000.

Close this Review now. Complete the changed retry independently and save both attempt histories.

R07 | DE04-07 | Functions and scope

Strong evidence shows

Different inputs produce expected results; returned value can be assigned/reused; no global variable is needed for the calculation.

Targeted repair direction

Check parameters -> local calculation -> return -> call site. Remove hidden globals.

Do not do

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Assigned changed retry

Write `shipment_cost(shipment_id, cartons, cost_per_carton)` returning a dict. Test three inputs including zero cartons.

Close this Review now. Complete the changed retry independently and save both attempt histories.

R08 | DE04-08 | Modules, imports and project structure

Strong evidence shows

Import succeeds from the documented folder; helper logic lives only in the helper module; running the entry file prints expected results.

Targeted repair direction

Run from the documented project folder. Check filenames/import names before touching sys.path.

Do not do

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Assigned changed retry

Create a fresh `shipment\_helpers.py` plus `run\_shipment\_demo.py` with different function names/inputs; do not copy your first file names.

Close this Review now. Complete the changed retry independently and save both attempt histories.

R09 | DE04-09 | Exceptions and debugging basics

Strong evidence shows

First error is preserved; each fix is isolated; expected bad input has a useful message; no bare/broad swallowing of exceptions.

Targeted repair direction

Let the failure reveal its exception type. Catch only expected errors; never `except: pass`.

Do not do

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Assigned changed retry

Write `parse_unit_cost(raw)` accepting numeric text such as "425.50" and rejecting "unknown" with a specific, visible error path.

Close this Review now. Complete the changed retry independently and save both attempt histories.

R10 | DE04-10 | Files, paths, CSV and JSON I/O

Strong evidence shows

Path does not depend on a random working folder; UTF-8 is explicit; input raw file remains unchanged; JSON read-back matches what was written.

Targeted repair direction

Print resolved paths; preserve raw files; use explicit UTF-8; validate structured read-back.

Do not do

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Assigned changed retry

Create a fresh manifest for `M04\_RETEST\_SHIPMENTS\_CANONICAL.csv`, then read its CSV header and validate the required fields without pandas.

Close this Review now. Complete the changed retry independently and save both attempt histories.

R11 | DE04-11 | pandas foundations: load, inspect, filter and aggregate

Strong evidence shows

Canonical controls: input rows=25; Completed rows=23; Completed sales=39320.00; missing sales_rep=1. Regional Completed totals: East 11700, North 9310, South 10210, West 8100.

Targeted repair direction

Audit in stages: input rows -> filter rows -> calculated field -> group grain -> independent total.

Do not do

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Assigned changed retry

Using the shipments retest CSV, calculate shipment_value, keep Delivered only, aggregate by hub, list top 3 Delivered shipments and count missing coordinator values. Use the separate retest controls only after your query is complete.

Close this Review now. Complete the changed retry independently and save both attempt histories.

R12 | DE04-12 | Practical Big-O intuition

Strong evidence shows

Explanation correctly distinguishes list scan from average set lookup and labels a simple nested full scan as $O(n^2)$.

Targeted repair direction

Count how the repeated work grows. Big-O is growth, not seconds. Prefer set/dict membership for repeated lookup.

Do not do

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Assigned changed retry

Compare "for each shipment, scan every known failed ID" with building a failed-ID set once. Explain the growth of both approaches.

Close this Review now. Complete the changed retry independently and save both attempt histories.

R13 | DE04-13 | Coding drills: lists and strings

Strong evidence shows

Run ``M04_CODING_DRILLS_VISIBLE_TESTS.py``; supplied tests pass and you add at least one edge case of your own.

Targeted repair direction

Write tiny input/output examples, shrink failing cases, inspect intermediate strings/lists, then rerun visible tests.

Do not do

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Assigned changed retry

In ``M04_FRESH_RETEST_TODO.py``, implement a changed ``normalize_words(text)`` task that strips extra spaces, lowercases tokens and returns them as a list. Add tests for blank and multi-space text.

Close this Review now. Complete the changed retry independently and save both attempt histories.

R14 | DE04-14 | Coding drills: dictionaries and sets

Strong evidence shows

Visible tests pass; output contract is deterministic; repeated duplicates are handled exactly as stated.

Targeted repair direction

Define duplicate/counting contract first. Dict for key totals; set for membership/uniqueness; test repeated duplicates.

Do not do

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Assigned changed retry

Implement ``count_statuses`` and ``unique_repeated_batch_ids`` on changed warehouse inputs inside the fresh-retest workspace.

Close this Review now. Complete the changed retry independently and save both attempt histories.

R15 | DE04-15 | Stack, queue, search and sort basics

Strong evidence shows

LIFO/FIFO outputs are correct; original list is preserved when required; linear search returns index/-1; binary search precondition is stated.

Targeted repair direction

Act out LIFO/FIFO with three labels. Check sorted-data precondition and whether sorting mutates the source.

Do not do

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Assigned changed retry

Use tasks ["ingest","validate","publish"] for stack/queue contrasts and search target 12 inside [4,12,7,3]. Sort a copy only.

Close this Review now. Complete the changed retry independently and save both attempt histories.

R16 | DE04-16 | Integrated Python checkpoint and reproducible script habits

Strong evidence shows

Clean run succeeds; controls reconcile; bad/missing input fails visibly; raw input is unchanged; output is deterministic; documented run command works from a fresh terminal.

Targeted repair direction

Split the checkpoint by boundaries: path -> parse -> schema -> calculation -> filter -> aggregation -> output -> rerun. Repair only the failed boundary.

Do not do

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Assigned changed retry

If assigned, complete `M04\_FRESH\_CHECKPOINT\_TODO.py` on the changed shipments fixture with R16 closed; preserve the original attempt.

Close this Review now. Complete the changed retry independently and save both attempt histories.